

Response Letter with Annotations

Mukerjee, S., Majó-Vázquez, S. & González-Bailón, S. (2018). Response to Webster and Taneja's Response to "Networks of Audience Overlap in the Consumption of Digital News." *Journal of Communication*, 68(1), 26–50. <https://doi.org/10.1093/joc/jqx007>.

Annotations

1. Purposes

- a. **Motivation (one sentence summary)** – This is a counter-response to a response about the original article. The motivation is to defend and clarify the methodological choices made in the original article against Webster & Taneja's critiques, particularly regarding significance testing, tie weighting, and conclusions about "massive overlap" in digital news consumption.
- b. **Specific Rationale (several bullet points)**
 - i. Webster and Taneja's comments lack a satisfactory benchmark for assessing significant audience overlap between outlets of vastly different reach
 - ii. Ignoring the strength of audience overlap obscures critical information about how audiences actually consume news
 - iii. Once overlap strength is accounted for, the data does not support the "massive overlap culture" conclusion
 - iv. Factual inaccuracies in Webster and Taneja's response needed correction
- c. **Research Questions** – Not an empirical study, no formal research questions

2. Methods

- a. **Research Design** – Commentary/rebuttal; defends methods from the original quantitative network analysis study
- b. **Sample (representative sample?)** – Sample from original study was comScore panel data on U.S. digital news audiences (panel size between 200,000 - 270,000 users; not 1 million as Webster and Taneja claimed); not a traditional representative sample but a large proprietary panel
- c. **Measures/Analysis**
 - i. Weighted network ties based on audience overlap between news outlets
 - ii. Confidence intervals around overlap estimates to assess statistical significance
 - iii. Weighted eigenvector centrality to rank outlet importance in the network
 - iv. Comparison of weighted vs. unweighted network conclusions

3. Results and Conclusions

- a. **Answers to RQs and Hypotheses**
 - i. Significance testing is appropriate even with population-level estimates; measurement error and stochastic variability justify modeling uncertainty

- ii. Weighted network analysis reveals that centrality rankings change *drastically* compared to unweighted analysis
- iii. The data reveals a core-periphery structure: a small core of highly accessed sites with heavy overlap, and a large periphery of niche outlets with loose connections
- iv. “Massive overlap” only exists at the core – it does not characterize overall news consumption behavior

b. Author’s Conclusions

- i. Dichotomizing ties (treating all connections as equal) is analogous to treating local roads and highways as equivalent – technically connected but misleadingly so
- ii. The observed patterns are more consistent with niche consumption theories than with a universal “massive overlap culture”
- iii. Methodological transparency and rigor are essential for valid theoretical conclusions

4. Discussion

a. Implications

- i. Network analysis of audience must account for tie strength to avoid misleading conclusions about media diets and exposure
- ii. The core-periphery finding has implications for theories of fragmentation, echo chambers, and niche media consumption
- iii. Statistical uncertainty in comScore (and similar) data should always be explicitly addressed

b. Limitations

- i. Analysis is limited to news sites only – findings may not generalize to other online media types
- ii. Reliance on comScore panel data, which carries inherent measurement uncertainty
- iii. Only one panel (one “sample” of possible panels) is available for analysis

c. Future directions

- i. Testing the impact of different thresholding techniques on audience network structure
- ii. Developing standardized measures that allow cross-country and longitudinal comparisons of audience network patterns
- iii. Broader goal of improving methodological transparency in audience research to enable cumulative theoretical progress

Response Letter

This study is a commentary and a rebuttal by the authors to defend their original study on digital news audience overlap networks against criticism from Webster and Taneja. At its core, the exchange is a methodological debate about how to properly construct and interpret audience networks using large-scale digital trace data.

The authors defend three central claims. The first is that significance testing is necessary even when working with population-level estimates from comScore, because measurement uncertainty and stochastic variability in the data-generating process justify the use of confidence intervals.

Second, the authors argue that ignoring the weight of audience overlap ties (i.e. treating all connections between news outlets as a binary of present or absent) produces misleading network structures and centrality rankings.

Finally, once tie strength is properly incorporated, the data reveals a core-periphery structure rather than the massive overlap culture which Webster and Taneja had previously posted as the proper analysis. In other words, heavy overlap only exists among a small cluster of high-reach outlets, while niche news sources remain loosely connected to this core.

The debate ultimately centers on the choices that are foundational to any network analysis of behavioral data: how to define edges, whether to threshold them, and how to handle uncertainty in large-scale observational datasets. It is a good reminder that methodological decisions are never neutral, and that transparency about those choices is essential for valid academic inference.

The authors make a compelling case for the strengths of their methodology, and that weighted network analysis provides substantially more nuanced and accurate insights than binary approaches, and they support this by demonstrating how centrality rankings shift dramatically when weights are included. Additionally, the core-periphery finding is theoretically meaningful and connects the empirical results of the study for the existing debate about media fragmentation and niche consumption.

The authors acknowledge in the paper that the analysis is limited to news sites, which constrains generalizability outside of that context. Audience overlap patterns across social media for example would likely behave differently. Being that this rebuttal is mostly defensive, the article does not substantively advance any new evidence.

As this article is essentially a smaller piece of a larger debate, there are plenty of questions to be taken from it, which I propose for further discussion surrounding the methodology and how researchers make methodological choices.

1. The authors argue that methodological choices are never arbitrary or neutral. How should computational methods scholars in communication develop shared standards for network construction in audience research, and who should set them? Is this a good idea or limiting to research?
2. comScore and similar datasets are widely used in computational communication research, but carry substantial measurement uncertainty, which Taneja and Webster took issue with.

How should researchers communicate these limitations to audiences who may treat large datasets as inherently reliable?

3. The authors propose developing standardized measures for cross-country and longitudinal comparison of audience networks. What methodological challenges would this pose, as related to the debate occurring in the current article. Is full comparability even possible given platform differences and different media environments?

